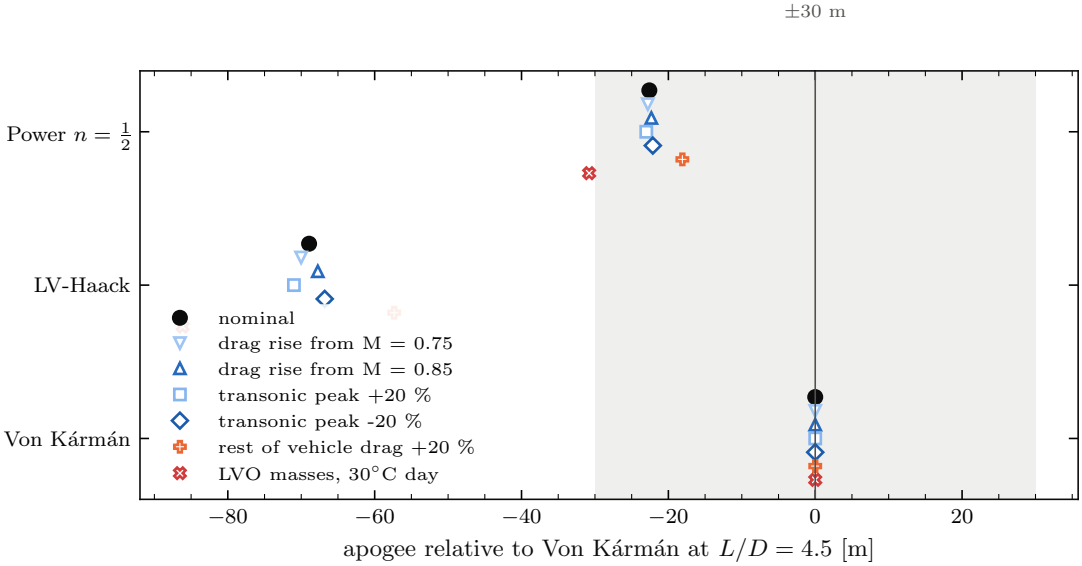

±30 m
Power $n = \frac{1}{2}$
LV-Haack
Von Kármán
nominal
drag rise from M = 0.75
drag rise from M = 0.85
transonic peak +20 %
transonic peak -20 %
rest of vehicle drag +20 %
LVO masses, 30°C day
−80
−60
−40
−20
0
20
apogee relative to Von Kármán at $L/D = 4.5$ [m]